THE ULTIMATE CHORD MATRIX

A quick and simple way to understand traditional harmony and compose great music.

Henrique Eisenmann
& Tim Lienhard

JAZZ LAB MUSIC

THE MATRIX



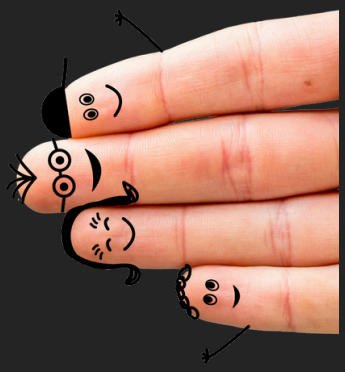
The chord Matrix is a method of organizing chords in a way that helps us visualize and understand harmonic progressions.

By using the concept of "Chord Family" and a system of arrows, it is possible to quickly analyze and recognize harmonic patterns in the music we love, especially Jazz/Pop/Rock.

In addition, we can use the matrix as a powerful composition tool, allowing one to create chord progressions that make sense and constructing forms that feel very natural.

Needless to say, this is one tool amongst millions of great theoretical devices. It doesn't replace a deep study of harmony, but allows one to quickly grasp complex concepts.

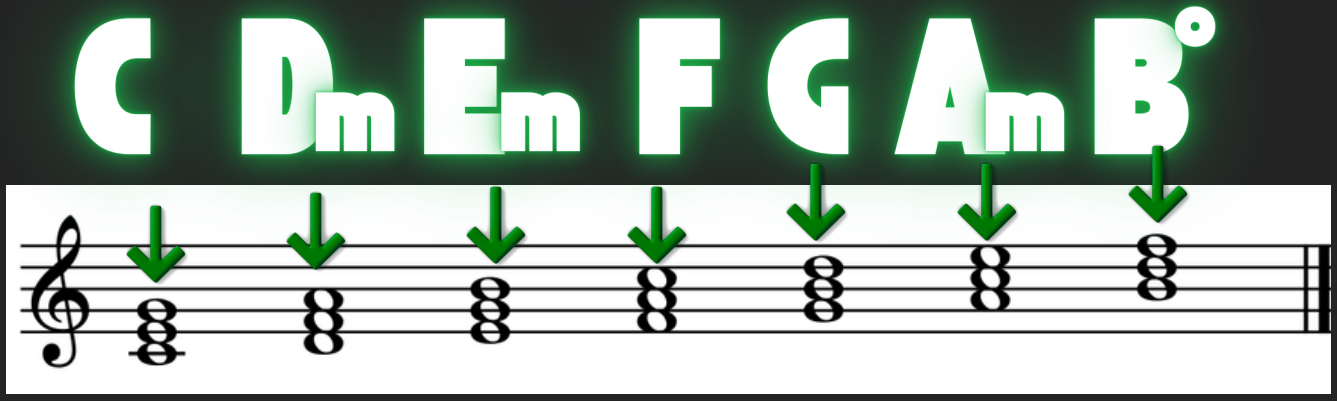
THE CHORD FAMILY



Traditional music theory calls them Diatonic Chords. We prefer to call them Chord Family, because the name implies relationships and hierarchies.

The chord family is a set of chords that are all born exclusively from the notes of a give scale. We normally start with triads (three-note chords with notes spaced in thirds) and move them through all notes of the scale.

The C major family looks like this:



The majority of pop songs nowadays utilize exclusively chords from the major family. Take a look at the A section of Beatle's "Let it Be":

C | G | Am | F
C | G | F | C

Notice how all chords above belong to the C major family. Try composing a simple song using only chords from the major family.

The B° is rarely used in this context. I recommend avoiding it. (vii° chords were originally a form of dominant chord that fell out of fashion in the 20th century). You still find diminisehd chords, but with different functions.

THE EXTENDED FAMILY



Let's expand our family in the Chord Matrix. We will now add the dominant chord for each of the members of our original C major family. These are the equivalent of the uncles/aunts/cousins of your original chords. A dominant chord is a major chord with a minor 7th a perfect fourth below the target chord.



In traditional music theory, these are called secondary dominants. Notice that they don't respect the original key signature - they may contains sharps or flats. Whenever a secondary dominant is used, our ears expect the next chord to be the resolution (where the arrow points). Whenever a composer breaks this expectation, we experience a deceptive cadence.
Let's take a look at Beatles' "Yesterday" (transposed to C major).

C | **E7** | Am | F G

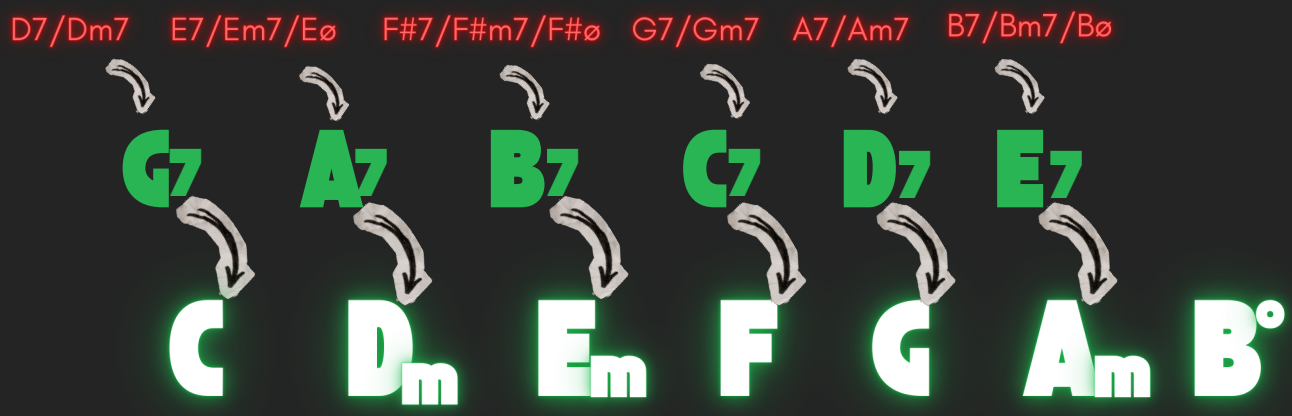
C | Am **D7** | F C

here the expectation is broken!
Our ears expected a G chord.

THE DISTANT COUSINS



We can expand our Matrix even more by adding the dominants of the dominants. These are like secondary degree cousins. Things can get a little complicated here. We will also add to the mix a set of minor chords that can also lead to the secondary dominants creating ii-V-I progressions. When the target chord in the nuclear family is minor, we can also include half dinished chords!



This may sound overly complicated to some, but it is important to say that this matrix was created based on usage and common practices, not theoretical elucubrations. Throughout history, more chords started to be used, and others fell out of fashion.

Whenever you see ond of the red chords, you are likely going to see a progression that follows the arrows. Follow the example below from "Fly me to the Moon"

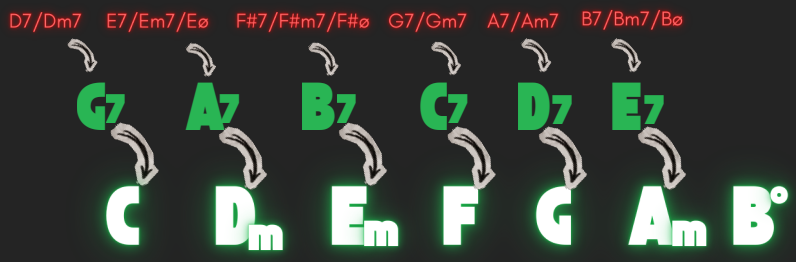
Am | Dm | G7 | C **C7**

F | **Bø** | **E7** | Am

THE UPSIDE DOWN WORLD



One more layer we can add to our Matrix is the inclusion the C minor chord family (the parallel minor scale to C major - always the natural minor scale). This operates like a darker upside down world (you must watch Stranger Things). Composers often use a couple of chords from the minor family, in a process known as "modal interchange". Whenever we use one of these chords, you can feel immediately hear the change, the progression becomes magical, mysterious, darker.



Cm D° Eb Fm Gm Ab Bb

Take a look at Coldplay's "Clocks". The Gm chord comes from the upside down world. That contributes to the magical feeling of this chord progression.

C | Gm | Gm | Dm

WHAT'S THE POINT OF ALL THIS?



- Having a blueprint to create succesfull chord progressions.
- Understanding the thought process behind many jazz standards and pop/rock songs.
- Visualizing the chords can help memorizing harmonic pathways, facilitating transpositions.

PRACTICE SUGGESTIONS

- Composing 8-bar long progressions
- Transposing and writing the chord matrix in all 12 keys.
- Analyzing Jazz repertoire, and trying to identify the hamonic progressions

This is not the only way to create chord progressions. Actually, music would be quite boring if we all just followed these templates. There are many other techniques out there: voice-leading, polychords, substitutions, modal progressions, atonal chords....

However, it is also important to understand the basic structures of jazz standards and common pieces, to develop fluency and control over musical languages.

If you are inspired, followed your intuition, not your theory blueprint.
But if you are not inspired (and very often we are not), it's extremely halpful to follow a template as a starting point, and then break some rules, add your own twist.

The next page contains a great exercise to compose a beatiful and effective harmonic template.

THE CHORD MATRIX



COMPOSING A JAZZ BALLAD TEMPLATE

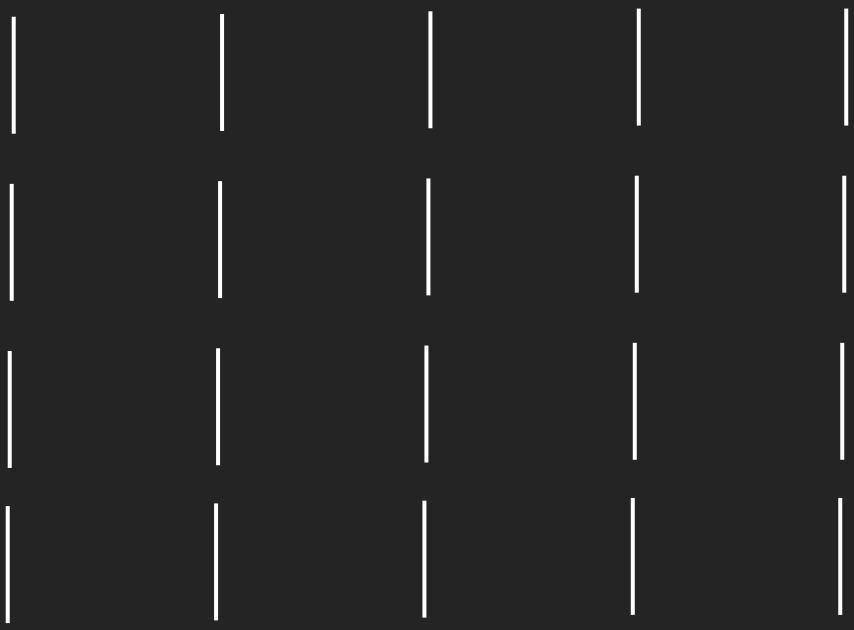
STEP 1: Start with C or Am. Fill up all the bars of the 16-bar template below with chords from the matrix. It is ok to choose randomly; often that's how you start the process. (If you want to create more of a pop song structure, copy bars 1-4 into bars 5-8).

STEP 2: Follow the arrows. If you use a secondary dominant, you must resolve in the direction of the arrow. (For Example, if you use a F#7, the next chord must be a B7 and the next one must be a Em).

STEP 3: If using a red followed by a green chord, feel free to include both in the same bar. (no need to hold each one for an entire bar).

STEP 4: If using chords from the minor family, be economic. If you use too many of them, the magic is lost.

STEP 5: Find a good balance between simplicity and complexity. Use mostly chords from the major family. Remember that if you mix too many colors together the result is brown.



If you are unhappy with the results, use your intutition and ear to identify bars that may need a different chord. If you know how to add more tensions (7ths, 9ths, 13ths, etc), feel free to enhance your chords.

i recommend writing 10-20 different compositions like this, until you start to identify specific patterns and sequences that you like.

The next step is to add a melody: use your voice to improvise some phrases and lines; you may also use target notes: one note per bar with ornamentations surrounding it.

Feel free to post your results and tag us on Instagram!

www.jazzlabmusic.org